higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING GRAPHICS AND DESIGN
(Second Paper)
NQF LEVEL 3

(6021023)

15 March 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: A2 drawing sheet

Drawing instruments may be used.

This question paper consists of 4 pages and 5 diagram sheets.

NCV

Copyright reserved

TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use both sides of the DRAWING SHEET.
5. Draw a 20 mm border on both sides of the DRAWING SHEET.
6. Do ALL drawings, including the candidate's information, in pencil.
7. A radius curve stencil may be used to draw arcs.
8. Use your own discretion where dimensions are NOT given.
9. ALL drawing work must conform to the latest SABS 0111 Code of Practice for Engineering Drawing.
10. A balanced layout is important and candidates will be penalised for poor planning.
11. Print neatly and legibly.

Copyright reserved

QUESTION 1

1.1 FIGURE 1, DIAGRAM SHEET 1 (attached), shows a table. Redraw the table and print the answers to the following questions using 3 mm lettering. (2)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and print only 'TRUE' or 'FALSE' next to the question number (1.2.1–1.2.4) in the table.

1.2.1 An isometric scale reduces the actual size of an object by 20%.

1.2.2 Ribs, webs, lugs, gear teeth and other thin features have NO section lines when the cutting plane passes at a right angle to the feature.

1.2.3 In the construction of a hexagonal bolt head, the measurement across the flats is 1,2 D where D is the nominal diameter.

1.2.4 A parts list is an index of the components that make up an assembly or a system.

(2 × 4) (8)

[10]

QUESTION 2

FIGURE 2, DIAGRAM SHEET 2 (attached), shows TWO views of the letter E in first-angle orthographic projection.

Do NOT copy the given views.

2.1 Construct an isometric scale. (2)

2.2 Draw an isometric view of the letter E using the isometric scale in first-angle orthographic projection. A must be the lowest point of the isometric drawing. (13)

[15]

QUESTION 3

FIGURE 3, DIAGRAM SHEET 3 (attached), shows a round pipe with a double angle. The pipe is open on both ends.

3.1 Draw, to scale 1:1, the given views. (4)

3.2 Develop the surface of the pipe. (13)

[17]

Copyright reserved

QUESTION 4

FIGURE 4, DIAGRAM SHEET 4 (attached), shows TWO views of a bearing mounting in first-angle orthographic projection.

Draw, to scale 1:1, the following detail drawings:

4.1 A full-sectional front view of item 1 (14)

4.2 A full-sectional front view of item 2 (6)

4.3 A top view of item 2 (4)
[24]

QUESTION 5

FIGURE 5, DIAGRAM SHEET 5 (attached), shows the components of a pulley assembly in third-angle orthographic projection. The parts list is as follows:

ITEM	NAME	NO PER ASSY
1	Bracket	1
2	Pulley	1
3	Bush	1
4	Shaft	1
5	Grub screw	1
6	Washer	1
7	M24 nut	1

5.1 Draw, to scale 1:1, an assembly drawing of a full-sectional front view. (27)

5.2 Insert the item numbers on the assembly drawing. (5)

5.3 Insert the title and scale centred beneath the assembly drawing. (2)
[34]

TOTAL: 100

Copyright reserved

DIAGRAM SHEET 1

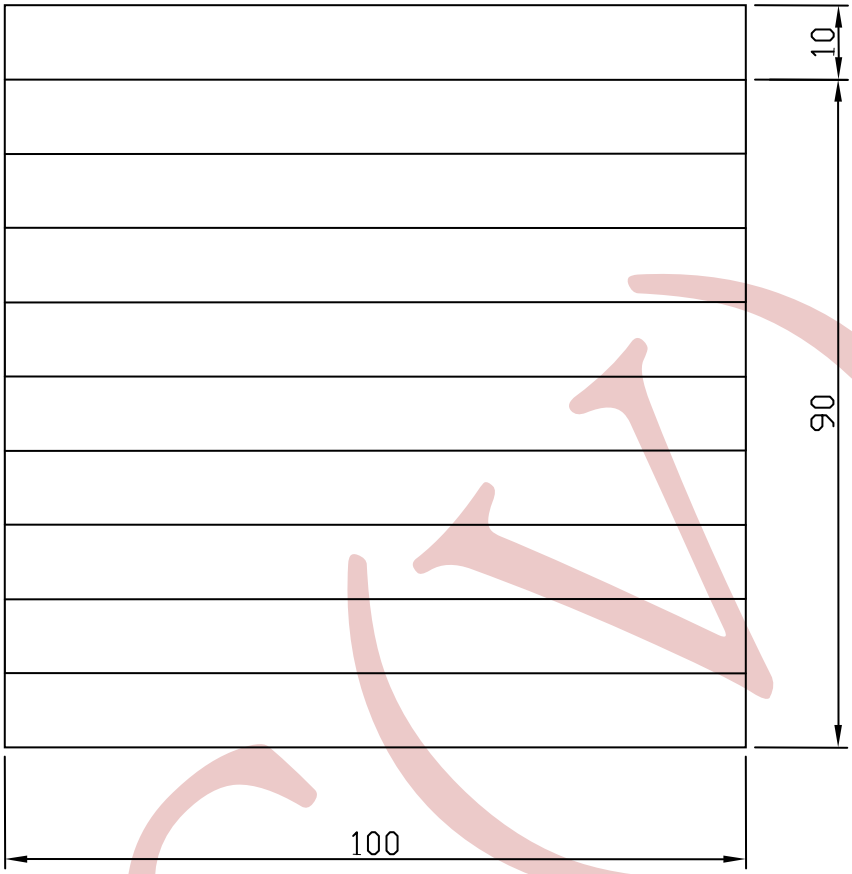


FIGURE 1

NCV

DIAGRAM SHEET 2

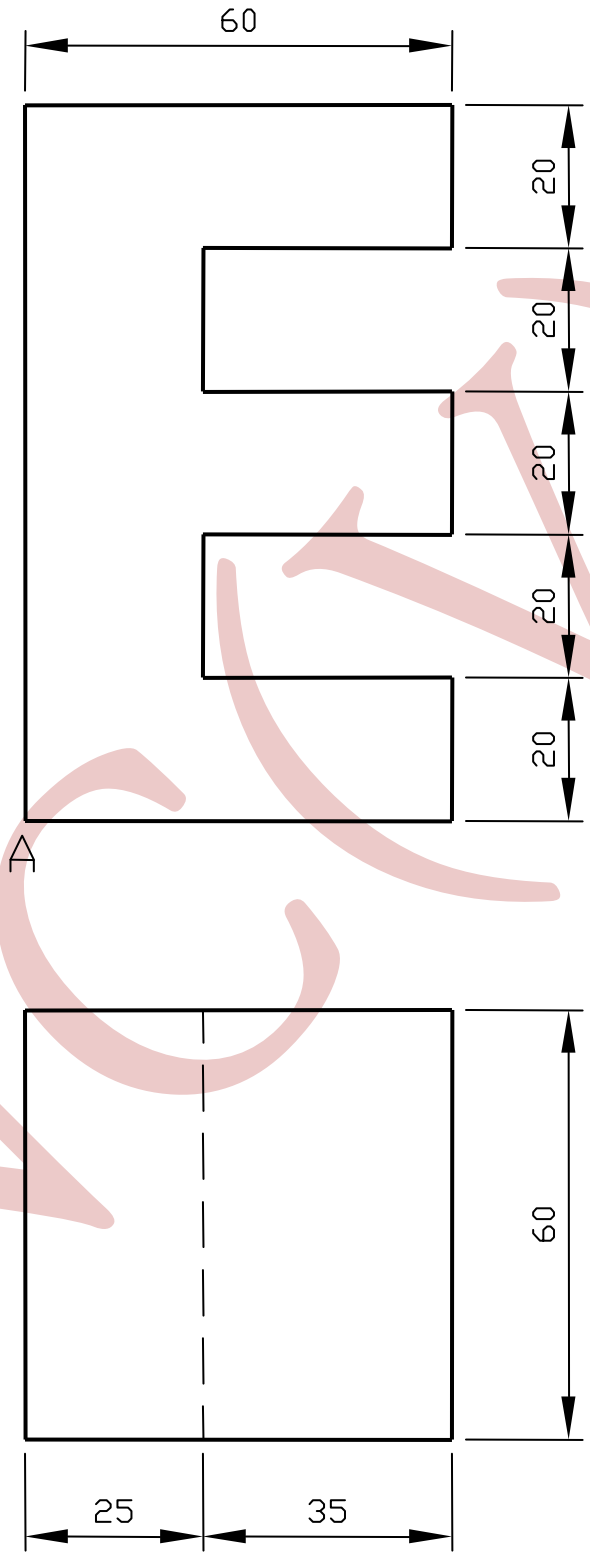


FIGURE 2

Copyright reserved

DIAGRAM SHEET 3

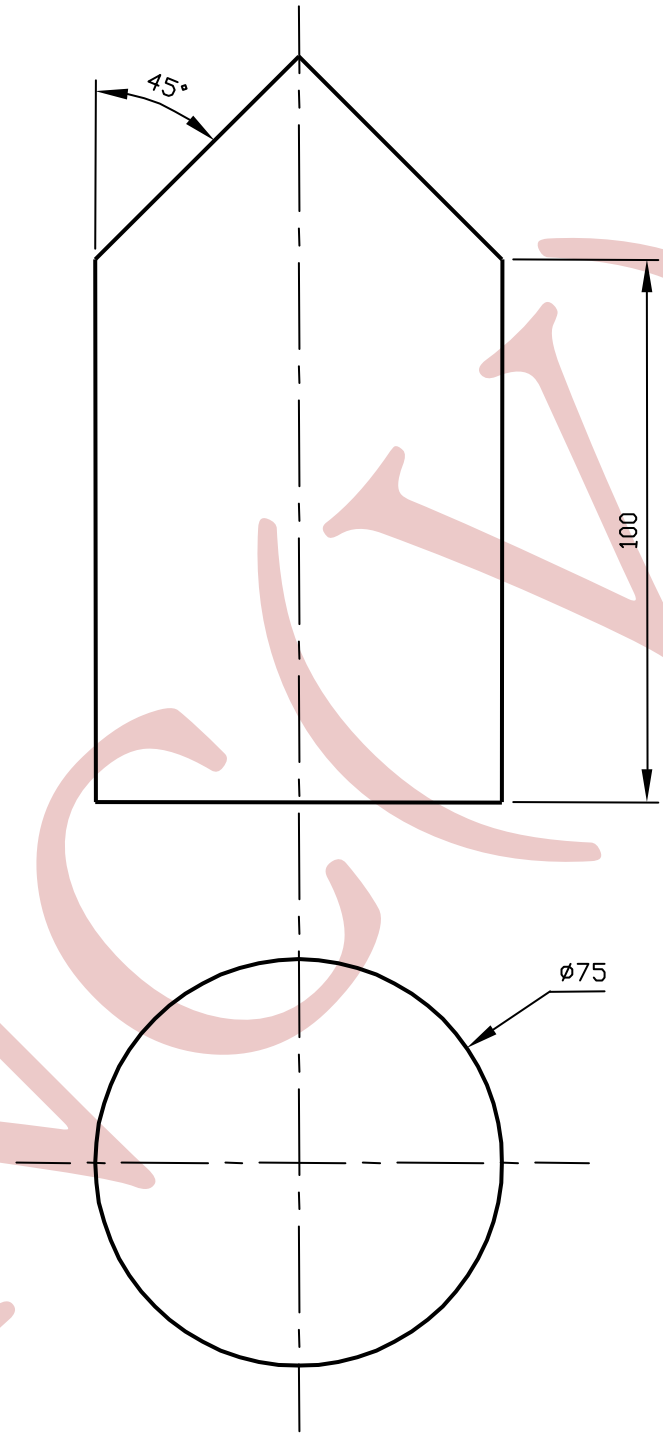


FIGURE 3

NCV

DIAGRAM SHEET 4

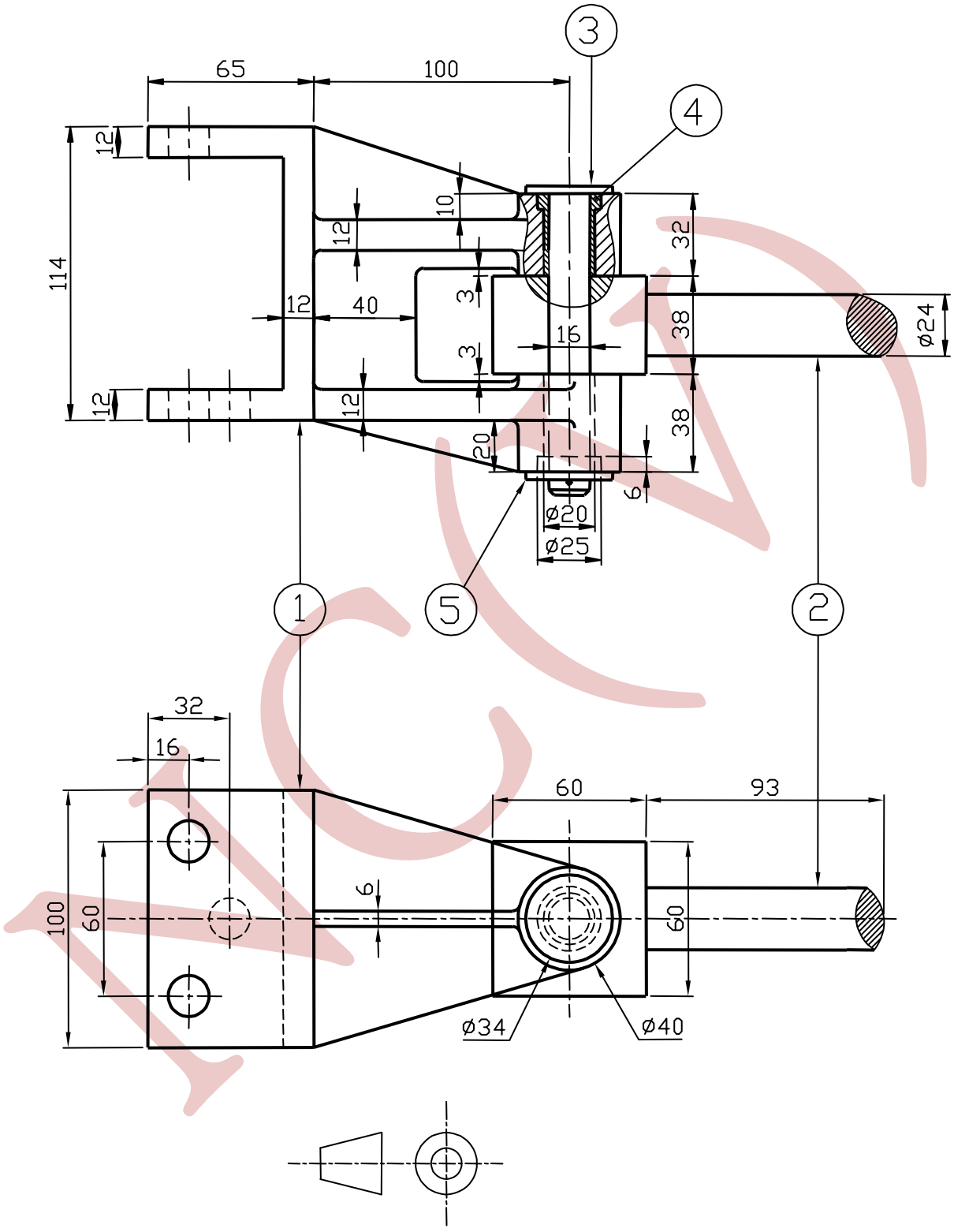


FIGURE 4

Copyright reserved

DIAGRAM SHEET 5

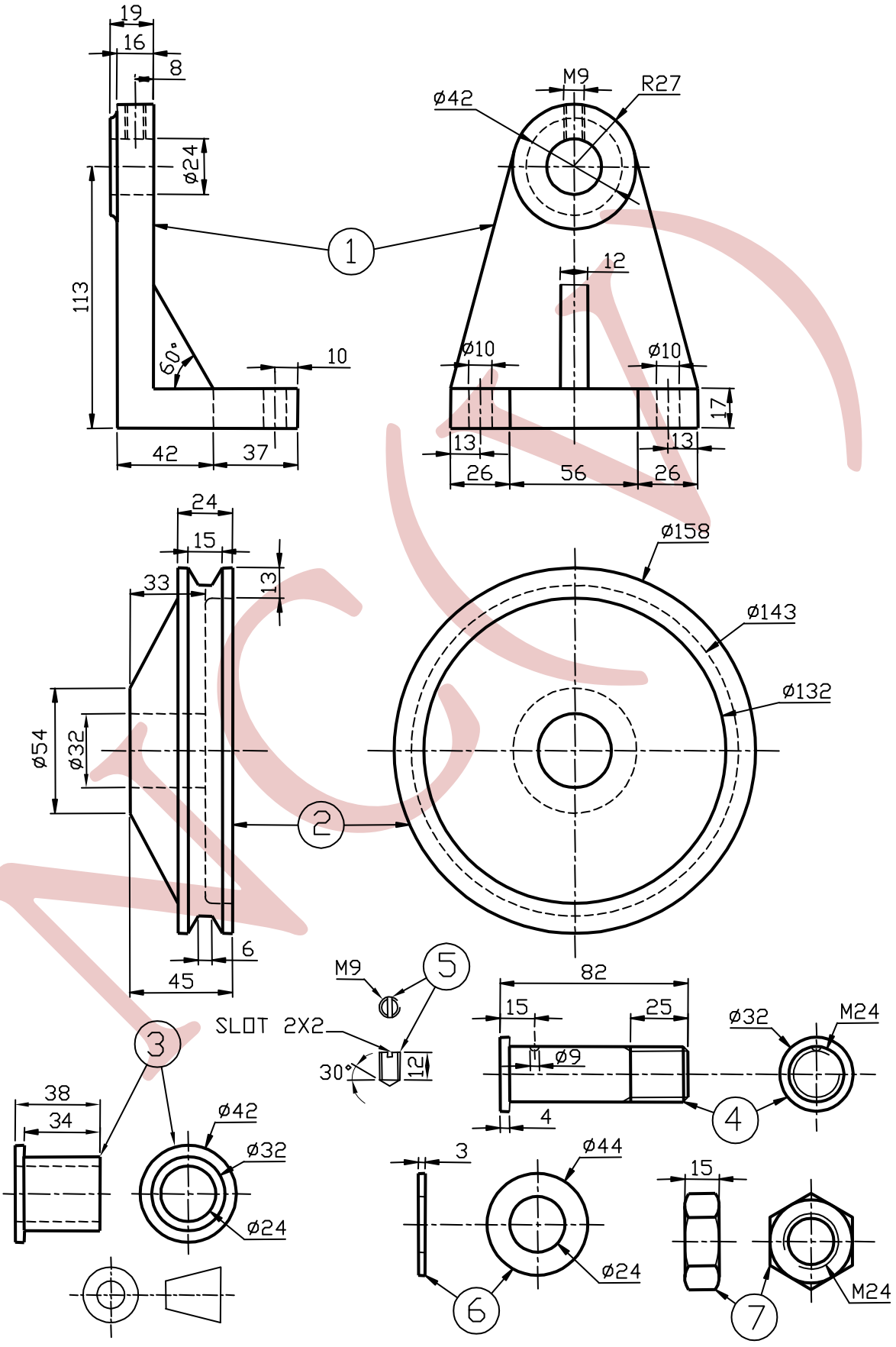


FIGURE 5

Copyright reserved